

AHAAN KOTHARI

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EDUCATION

Bachelor of Science, Mechanical Engineering

May 2026

Rose-Hulman Institute of Technology – Terre Haute, IN

GPA: 3.67

Relevant Courses: Data Structures and Algorithms, Objected Oriented Software Development, Web Programming

Dean's List Scholar every quarter

INTERNSHIP EXPERIENCE

Dalal Engineering Pvt. Ltd. | *Engineering Intern* | Mumbai, India

June – Aug. 2025

- Authored comprehensive technical documentation for PLC operation, safety protocols, and on-boarding, reducing training time for new operators by 20%.
- Conducted quality control and troubleshooting on Sedomat 6000 PLC systems, improving process uptime and reducing errors in package dyeing operations
- Researched and recommended cost-effective safety optimizations, projected to lower maintenance costs by 15% annually for safety features annually

Riverstone Jewels | *Engineering Consultant* | Mumbai, India

June – Aug. 2025

- Developed and implemented a centralized database for CAD files, sketches, and design intent, enhancing team accessibility and reducing file retrieval time by 40%
- Analyzed pre-casting workflows to identify automation opportunities and proposed process enhancements for mold setting and machining, cutting manual steps by 25% and improving operational efficiency

Doctor and Doctor Law Firm | *Law Intern* | Mumbai, India

Jan. - March 2025

- Supported civil litigation activities under Judge Arif Doctor, conducting due legal research and case analysis
- Reviewed civil statutes and precedents to support case preparation
- Attended court hearings and depositions to observe civil law procedures

PROJECT EXPERIENCE

Maudieai.com | *Founder & Solo Developer*

Feb. 2026- Present

- Architected a full-stack AI platform automating FDA submission documents for medical device manufacturers
- Engineered a multi-agent RAG pipeline combining Next.js, React, PostgreSQL and Mistral 7B LLM to generate compliance submission draft with HITL workflows.
- Focus: Building a platform medical device companies trust with sensitive compliance work by combining security-first architecture, error handling and audit trails.

QuickBurst Speed-training | MDS410 – Multidisciplinary Capstone

Sept. - May 2026

- Designed a portable, battery-powered speed training system with undirected resistance, sprint feedback, and motorized over-speed assistance
- Integrated control algorithms, sensor suite, and wireless app interface, achieving sub-0.01 s timing accuracy while maintaining biomechanically safe loads

Cummins Xtern Challenge | *AI/ML Engineer* | Columbus, IN

Feb. - March 2026

- Placed 2nd in 3-week hackathon to build an AI-powered diagnostic tool for Cummins field service technicians
- Implemented a multi-agent RAG system (**FieldAssist**) using Mistral 7B, LangGraph, and ChromaDB vector store to process SAE fault codes for contextual repair guidance.
- Presented technical solution and business case to Cummins leadership including CEO; produced documentation covering architecture, governance, and regulatory compliance.

StudySync Web Application | CSSE280 – Intro to Web Programming

June – Aug. 2025

- Developed a secure campus study-session platform with Firebase authentication, REST APIs, and real-time chat/file sharing via Socket.io

Finite Element Analysis (FEA) Projects | ME480 – Machine Component Design

May - June 2025

- Modeled axially-loaded shaft with keyway in SolidWorks; computed stress concentration factors and validated results against theoretical predictions

Smart Hangboard Extension | ME430 – Mechatronics

Sept. – Nov. 2025

- Engineered Arduino-based climbing module integrating load cell, HX711 amplifier, and IR sensors to capture real-time force data

- Built servo-actuated LCD platform with fully integrated sensors in a professional-grade assembly
- Hydrocycle Project** | Summer 2024 Project **June – Aug. 2024**
- Designed and fabricated water-capable bicycle; achieved 200% load capacity increase after iteration redesign
 - Constructed entirely from recycled materials, applying sustainable engineering principles
- Design Build Test Project** | EM103 – Intro to Design **March - June 2024**
- Manufactured functional hurdy-gurdy using industrial tools, SolidWorks CAD, and additive manufacturing.
 - Used 99% recycled materials and performed instrument in campus concert, demonstrating craftsmanship and sustainability awareness
- Formula 1 Racecar Model** | EM104 – Graphical Communications **Sept. – Nov. 2023**
- Designed detailed 3D model in SolidWorks, demonstrating proficiency in surface modeling and complex assemblies
- Lightweight Linkage Design** | EM121 – Statics and Mechanics of Materials **Nov. – Feb. 2024**
- 3D-printed and tested load-bearing mechanical link for resistance-training device, optimized for minimal mass and space efficiency.
 - Evaluated materials using SolidWorks Simulation; awarded 2nd place in institute competition.
- Drone Security System** | Start-up Hackathon **Oct. – Nov. 2023**
- Co-founded concept integrating aerial surveillance with motion tracking for automated security responses
 - Conducted market research, user interviews, and stakeholder validation; securing \$2000 funding by winning 1st place
- Jetpack App** | CSSE220- Object-Oriented Software Development **Sept. – Nov. 2024**
- Developed Java-based game simulating Jetpack Joyride with sprite animation, sound control, and infinite scrolling.
 - Implemented object-oriented architecture with collision detection and event handling for real-time user response.
- Robotic Arm Sorter** | ME123 – Computer Programming **Feb. - March 2024**
- Programmed robotic arm to sort colored beads via sensor-based feedback loops.
 - Designed motion control logic to minimize travel distance using optimization techniques.
- Asteroid Landing System Simulation** | ME123 – Computer Programming **Feb. - March 2024**
- Developed predictive algorithm modeling asteroid re-entry trajectory under gravitational and atmospheric influences
 - Estimated landing coordinates within defined radius with high accuracy.

VOLUNTEER EXPERIENCE

- Alpha Phi Omega Service Fraternity** | Active Member | Terre Haute, IN **Nov. 2023 - Present**
- Contributed 40+ service hours annually through community initiatives and environmental restoration.
 - Assisted infrastructure rehabilitation for senior citizens through Wabash Valley Neighbors Helping Neighbors.
 - Planted and maintained dozens of trees in local communities via the Trees movement.
 - Supported elementary students' literacy development in Reading Boot Camp
 - Led a personal project to assemble and deliver a custom desktop computer for a student in need.

ACTIVITIES

Tutor at AskRose Homework Help	11/23 - Present
Grader for Numerical Methods	11/25 - Present
South Asian Student Alliance (SASA) Member	09/23 - Present
Alpha Phi Omega Service Fraternity	11/23 - Present
Triangle Fraternity Class Treasurer	10/23 - Present
Rock Climbing Club Member	10/24 - Present

SKILLS

Software: MATLAB, SolidWorks, C++, Java, Python, R, Maple, Microsoft Office, HTML, JavaScript, Simulink, LangGraph, ChromaDB, RAG (Retrieval-Augmented Generation), Mistral LLM, React, Node.js, PostgreSQL

Technical: CAD modeling and drafting, FEA, lathe, welding, additive manufacturing, instrumentation

Professional: Process optimization, product testing and validation, technical documentation

Languages: English, Spanish, conversational Hindi